

Research Notes on Independent component analysis

Independent component analysis

>> Section 1

where the A is an $m \times m$ invertible matrix called the mixing matrix, s_i represents the m -dimensional vector containing the values of the sources at time t_i , and x_i is the corresponding vector of observed values at time t_i . The goal is to estimate both A and the source signals $\{s_i\}$ solely from the observed data $\{x_i\}$. After centering, the Gram matrix is computed as:

>> Section 2

Based on maximum likelihood estimation Maximum likelihood estimation (MLE) is a standard statistical tool for finding parameter values (e.g. the unmixing matrix W) that provide the best fit of some data (e.g., the extracted signals y) to a given a model (e.g., the assumed joint probability density function (pdf) p_s of source signals). The ML "model" includes a specification of a pdf, which in this case is the pdf p_s of the unknown source signals s . Using ML ICA, the objective is to find an unmixing matrix that yields extracted signals $y = Wx$ with a joint pdf as similar as possible to the joint pdf p_s of the unknown source signals s . MLE is thus based on the assumption that if the model pdf p_s and the model parameters A are correct then a high probability should be obtained for the data x that were actually observed. Conversely, if A is far from the correct parameter values then a low probability of the observed data would be expected. Using MLE, we call the probability of the observed data for a given set of model parameter values (e.g., a pdf p_s and a matrix A) the likelihood of the model parameter values given the observed data. We define a likelihood function $L(W)$ of W :

>> Section 3

$$h(Y) = \frac{1}{N} \sum_{i=1}^M \sum_{t=1}^N \ln \left(1 - \tanh^2 \left(w_i^T x_t \right) \right) + \ln |W|$$

>> Section 4

Linear noiseless ICA The components x_i of the observed random vector $x = (x_1, \dots, x_m)^T$

are generated as a sum of the independent components s_k , $k = 1, \dots, n$:

>> Section 5

Linear noisy ICA With the added assumption of zero-mean and uncorrelated Gaussian noise $n \sim N(0, \text{diag}(\Sigma))$, the ICA model takes the form $x = As + n$.

>> Section 6

$G_1 = \frac{1}{a_1} \log \left(\cosh \left(a_1 u \right) \right)$ and $G_2 = -\exp \left(-u^2 \right)$

>> Section 7

General Derivation In the classical ICA model, it is assumed that the observed data $x_i \in R^m$ at time t_i is generated from source signals $s_i \in R^m$ via a linear transformation $x_i = As_i$, where A is an unknown, invertible mixing matrix. To recover the source signals, the data is first centered (zero mean), and then whitened so that the transformed data has unit covariance. This whitening reduces the problem from estimating a general matrix A to estimating an orthogonal matrix V , significantly simplifying the search for independent components. If the covariance matrix of the centered data is $\Sigma_x = AA^T$, then using the eigen-decomposition $\Sigma_x = QDQ^T$, the whitening transformation can be taken as $D^{-1/2}Q^T$. This step ensures that the recovered sources are uncorrelated and of unit variance, leaving only the task of rotating the whitened data to maximize statistical independence. This general derivation underlies many ICA algorithms and is foundational in understanding the ICA model.

>> Section 8

$$K = \frac{E \left[(y - \overline{y})^4 \right] (E \left[(y - \overline{y})^2 \right])^2 - 3 (E \left[(y - \overline{y})^2 \right])^3}{E \left[(y - \overline{y})^4 \right] (E \left[(y - \overline{y})^2 \right])^2 - 3 (E \left[(y - \overline{y})^2 \right])^3}$$

>> Section 9

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If we substitute a commonly used high-Kurtosis model pdf for the source signals $p_s = (1 - \tanh(s)^2)$ then we have

>> Section 10

The joint pdf p_Y can be shown to be related to the joint pdf p_y of the extracted signals by the multivariate form:

>> Section 11

$$\ln L(W) = \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \ln (1 - \tanh(w_i^T x_t)^2) + \ln |\det W|$$

>> Section 12

since $H(x) = -\frac{1}{N} \sum_{t=1}^T \ln p(x_t)$, and maximizing $L(W)$ does not affect $H(x)$, so we can maximize the function